

Inferring the past from the present phylogeographic structure of North American forest trees: seeing the forest for the genes

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Abstract: The study of past historical events that have led to ecological changes is a recurrent topic in many disciplines. Given that many of these events have left a large and long-lasting evolutionary imprint on the extant population genetic structure of species, phylogeographic studies on modern taxa have been largely used to infer the impacts of these events and to complement previous paleoecological and paleobotanical surveys. In spite of the geographical and geological complexity of North America, converging patterns can be observed when comparing the available genetic data for forest trees. Such patterns include the co-location of genetic discontinuities among species and their coincidence with mountain ranges (e.g., the Appalachians, the Rocky Mountains, the Sierra Nevada, or the Transverse Volcanic Belt) and with previously inferred glacial refugia. Using examples drawn from the available literature, we illustrate such shared features and present the contrasting phylogeographic patterns observed among the different regions of the continent. The various evolutionary consequences of historical events that can be deduced from these phylogeographic studies (e.g., past bottlenecks, founder effects, allopatric divergence, or introgressive hybridization) are additionally discussed. The present challenges and future research prospects that are likely to further advance this field are finally outlined.

Résumé : L'étude d'événements passés responsables de bouleversements écologiques est un sujet récurrent dans plusieurs disciplines. Puisque plusieurs de ces événements ont laissé une trace évolutive importante et durable dans la structure génétique des populations d'espèces, la phylogéographie des taxons contemporains a fréquemment été étudiée afin de déduire les impacts de ces événements et d'enrichir les études antérieures en paléoécologie et paléobotanique. Malgré la complexité géographique et géologique de l'Amérique du Nord, les auteurs ont noté des patrons convergents en comparant les données génétiques disponibles chez les arbres forestiers. Ces patrons incluent la présence de discontinuités génétiques communes aux espèces et leur coïncidence avec les chaînes de montagnes (p. ex., les Appalaches, les Rocheuses, la Sierra Nevada ou la Cordillère néovolcanique au Mexique) ainsi qu'avec les refuges glaciaires déduits antérieurement. À l'aide d'exemples tirés de la littérature, les auteurs illustrent de telles caractéristiques partagées et présentent les patrons phylogéographiques contrastés observés dans les différentes régions du continent. Les diverses conséquences évolutives des événements historiques qui peuvent être déduites des études phylogéographiques sont également discutées (p. ex., les goulots d'étranglement anciens, les effets fondateurs, la divergence allopatrique et l'hybridation introgressive). Enfin, les auteurs abordent les défis actuels et les avenues futures de recherche qui pourraient faire progresser le domaine.

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Introduction

The knowledge of past historical events and population dynamics has been a constant quest in evolutionary biology. Such knowledge plays a key role in understanding present phenomena and in predicting the outcomes of future environmental changes (e.g., Thomas et al. 2004; Hampe and Petit 2005; Provan and Bennett 2008). The population dy-

namics of a species is usually driven by both intrinsic features inherent to each taxon and external biotic and abiotic pressures (Hamrick and Godt 1996; Aguinagalde et al. 2005). The study of past plant population dynamics has traditionally been conducted by using geological and fossil data to reconstruct paleofloras over different periods of time (e.g., Davis 1983; Matthews et al. 1989). These paleofloras are thus used to infer the events that shaped their evolution-

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ary history (e.g., Jackson et al. 1997). However, despite the advances in the detection and simulation methods used in paleobotany (e.g., Jackson et al. 1997; Edwards et al. 2000), several constraints, such as the unavailability of data for some areas, the bias towards some specific taxa (e.g., wind-pollinated plants), and the low taxonomical and morphological resolution of most fossil remnants, can blur their conclusions (Cruzan and Templeton 2000). For example, in North America, the lack of morphological resolution among closely related species and (or) species complexes has restricted many of the pollen-based biome reconstitutions to the genus level (e.g., Davis 1983; Jackson et al. 1997). With trees, these constraints can be further amplified by the large plasticity in phenotypic characters often seen within widely distributed species (e.g., Richardson et al. 2000; Rehfeldt et al. 2001).

During the last 20 years, a few thousand studies investigating the distribution of genetic variation have been conducted in different organisms including plants, animals, bacteria, and fungi (Petit and Vendramin 2006). These studies, usually joint under the term “phylogeography”, address topics such as the location of origins, factors of vicariance, and patterns of migration of species. Although the strict definition of the term phylogeography assumes a deep knowledge of the phylogeny of intraspecific lineages, which is used to understand how the distribution of these lineages has been shaped by history (e.g., Avise 1989), this term is also used in a broader sense. This sense implies the detection of geographic patterns in the distribution of genetic variation without the use of a proper phylogenetic approach (i.e., genogeography; Manni et al. 2004). The comparative analysis of the patterns detected in these phylogeographic studies with those from palynological, paleobotanical, geographic, or geological surveys opens a new dimension in the study of vegetation dynamics and its association with past and future climate change (Hu et al. 2008; Petit et al. 2008). Among others, it allows the detection of converging patterns, such as the location of glacial refugia and postglacial recolonization routes, and the inference of common causes of vicariance at the regional scale (e.g., Brunsfeld et al. 2001; Carstens et al. 2005; Naydenov et al. 2007).

In Europe, phylogeographic studies have been performed in both animals and plants at the continental level (e.g., Hewitt 1996; Taberlet et al. 1998), while in North America, they have been mainly conducted at the regional scale (e.g., Avise and Walker 1998; Walker and Avise 1998) and almost exclusively in animal taxa (but see Soltis et al. 1997, 2006; Brunsfeld et al. 2001). This is especially due to both the high number of species and the complex and heterogeneous geographical and geological history of this continent, which makes any attempt to conduct large comparative and quantitative phylogeographic studies a tremendous task.

The present review represents a first attempt to assemble and interpret, at the continental scale, the main phylogeographic patterns for North American forest trees, as deduced from available population genetic studies. Forest trees are useful models to study the impacts of historical events due to their generally large population sizes, long generation times, high investment in reproduction, and high phenotypic plasticity, which allow them to survive drastic environmental changes and to preserve the genetic imprint of such

events for a long period (Hamrick 2004; Petit and Hampe 2006). Here, we briefly describe the rationale and the methods used to detect population genetic patterns related to the effects of historical events, we survey the general distribution of phylogeographic breaks in the continent, and we describe the potential evolutionary trends that can be retrieved from the available phylogeographic studies. Finally, we discuss the future challenges, methods, and prospected research opportunities in the field. Given that many of the subjects (i.e., hybridization, introgression, speciation, migration patterns, etc.) mentioned in this paper are large enough to each deserve their own review, they are only addressed in the context of a general survey. We thus encourage the readers interested in these subjects to refer to the cited works and references therein for a more in-depth treatment. Similarly, while the quantitative analysis of phylogeographic patterns has been possible within various regions of North America (e.g., Carstens et al. 2005; Soltis et al. 2006) by considering different taxonomical groups (i.e., forest trees, annual plants, animals, etc.), such analyses are currently out of scope at the continental scale for North American forest trees, given the insufficient and fragmented nature of the available data.

Population genetic structure and inference of phylogeographic patterns

The population genetic structure of a species refers to the way that its different genetic lineages are sorted in space and time. It is usually modeled by its different life history traits and by the evolutionary and historical events that it has had to cope with. In plants, different attempts have been made to correlate the estimates of genetic diversity and population structure with life history traits (e.g., Hamrick and Godt 1996; Nybom 2004; Aguinalgalde et al. 2005). Such comparative studies, based on the variation of nuclear and chloroplast genetic markers, have revealed that the amounts of diversity and population differentiation are related, at least partially, to some biological features such as abundance, longevity, or dispersal ability. For example, wind-pollinated and widespread plant taxa tend to have higher amounts of nuclear genetic diversity and to exhibit lower population differentiation than animal-pollinated and endemic species (Hamrick and Godt 1996). However, these life history and ecological traits only explained about 35% of the total nuclear genetic variation (Hamrick and Godt 1996; Nybom 2004). Historical factors inherent to each taxon were thus held responsible for the unexplained variation among taxa (e.g., Qiu and Parks 1994). For instance, in European angiosperms, the most critical factors explaining the distribution of their neutral chloroplast DNA (cpDNA) variation appeared to be the location and the degree of isolation of glacial populations (Aguinalgalde et al. 2005). In North America, the Pleistocene glaciations are also usually invoked to explain many of the phylogeographic patterns observed in different taxa including forest trees (e.g., Fournier and Adams 1986; Mitton et al. 2000; Godbout et al. 2005 among many others). However, other geological, ecological, and biological features have to be considered in the delineation of converging patterns such as in Europe (see Soltis et al. 1997; Carstens et al. 2005).

In seed plants, the spread of genes is mediated by two in-

dependent vehicles that operate asymmetrically: the pollen and the seed. These two vehicles, which are dispersed in a variety of forms, generally carry different kinds and amounts of genetic material. In most seed plants, the nuclear genome is biparentally transmitted and it is equally dispersed by pollen and seed. In contrast, the cytoplasmic genomes are inherited and dispersed in different ways (reviewed by Harris and Ingram 1991; Mogensen 1996; Petit and Vendramin 2006 among others). For example, in many angiosperms, both the cpDNA and the mitochondrial (mtDNA) genomes are maternally inherited (Harris and Ingram 1991; Mogensen 1996) and thus exclusively dispersed by seed, while in gymnosperms, the inheritance and way of dispersal of cytoplasmic genomes differ among families. For instance, in the Cupressaceae, both cytoplasmic genomes are paternally transmitted (e.g., Neale et al. 1991) and thus spread by both pollen and seed, while in the Pinaceae, mtDNA is maternally inherited and seed dispersed and cpDNA is paternally transmitted and distributed by both pollen and seed (e.g., Neale and Sederoff 1989).

Following different theoretical studies (e.g., Birky et al. 1989; Hu and Ennos 1997 among others), it is expected that population structure estimates determined with maternally inherited and seed-dispersed cytoplasmic markers are more prone to reveal the “genetic imprint” of past historic events than the estimates obtained from biparentally or paternally inherited markers. This is mainly for two reasons. First, haploid cytoplasmic genomes have an effective population size (N_e) that is one half or one quarter that of nuclear diploid markers, depending whether the species is monoecious or dioecious, respectively (Birky et al. 1989). Thus, given that genetic drift is a direct function of N_e , cytoplasmic genomes are expected to be more susceptible to the action of this stochastic force (Birky et al. 1989) and develop a more significant population structure over time than nuclear diploid genomes. Second, given that seeds have generally lower dispersal capabilities than pollen, markers of seed-dispersed genomes are expected to retain this population structure for a longer period than those of pollen-dispersed genomes due to more limited effects of gene flow. This observation was confirmed by extensive comparative studies in plants (e.g., Petit et al. 2005; Petit and Vendramin 2006) that revealed higher levels of population differentiation for maternally inherited and seed-dispersed markers than for paternally inherited and pollen-dispersed markers. Indeed, pollen-dispersed markers only exhibited such a structure when the amount of gene flow among populations since their initial isolation has been very low (e.g., Ritland et al. 2001; Irwin 2002; Mylecraine et al. 2004).

A number of methods have been developed to detect patterns of genetic variation that can be related to geography (reviewed by Storfer et al. 2007). The most commonly used are spatial (e.g., Dupanloup et al. 2002; Manni et al. 2004), multivariate (e.g., Cavalli-Sforza et al. 1994), Bayesian (e.g., Pritchard et al. 2000), and nested-clade analyses (e.g., Templeton 2001). The comparison of population differentiation indices that take (e.g., N_{ST} or R_{ST}) and do not take (G_{ST}) into account the relatedness among genetic variants (Pons and Petit 1996) has also become common (Petit et al. 2005). Additional procedures such as tests for bottlenecks (e.g., Luikart and Cornuet 1998), mismatch distribution

(Rogers and Harpending 1992), spatial autocorrelation (e.g., Slatkin and Arter 1991), or estimates of dispersal and migration rates (reviewed by Neigel 1997 among others) are also used to investigate further evolutionary and historical trends including past reductions of population size, founding events, inbreeding depletion, admixture (i.e., introgression), or speciation. The subsequent comparison of the results of these analyses with those from paleobotanical or palynological studies generally provides a reasonable picture of the different processes that have driven the modern population structure of a species or a group of species (e.g., Hu et al. 2008).

Genetic discontinuities in North American forest tree species

Genetic discontinuities are generally the consequence of past population isolation induced by historical events or by geographic and biological features that hampered the exchange of genes among the isolated populations (e.g., Avise 1989). The first common genetic discontinuities in North America were reported at the regional scale for aquatic taxa (e.g., Avise 1989; Walker and Avise 1998). Most of these phylogeographic breaks appeared to be correlated with the location of river basins. Some of these patterns and other additional discontinuities associated with mountain ranges also seemed to be present in annual plants and in forest trees (e.g., Soltis et al. 1997, 2006).

Considering the high ecological and geological heterogeneity and complexity of North America, we have subdivided the continent into five large geographic regions to describe the main genetic discontinuities reported to date for forest trees: eastern North America (from the Great Lakes region and the coasts of New England to the Gulf coasts and Florida including the central plains and steppes), boreal North America (from the Maritime Provinces of Canada to Alaska and from the Central Plains to the arctic tree line including the cryptic northern refugia), the Pacific Northwest (from the northern coasts of California to the southern coasts of Alaska and east to the Rocky Mountains), the Californian Floristic Province (California and northern Baja California), and northern and central Mexico (from the Sonora Desert to the Isthmus of Tehuantepec).

Eastern North America

A number of genetic discontinuities reported in this region for freshwater fish (e.g., Avise 1989), aquatic and terrestrial turtles (Walker and Avise 1998), annual plants (e.g., Maskas and Cruzan 2000; Griffin and Barret 2004), and some forest trees (Table 1) appear to be located between the glacial refugia inferred by palynological surveys (reviewed by Soltis et al. 2006). These refuges include the plains at both the Gulf and Atlantic coasts and central Florida (e.g., Davis 1983; Jackson et al. 1997). For example, the study of cpDNA restriction sites and allozymes in *Liriodendron tulipifera* showed that the populations from central Florida bear a distinct genetic background from that of the northern and coastal stands (Fig. 1A) (Parks et al. 1994; Sewell et al. 1996). Such a subdivision was also observed in *Acer rubrum* with cpDNA sequences (McLachlan et al. 2005). In *Fagus grandifolia*, a study of nuclear allozymes

Table 1. Number of genetically distinct groups and distribution patterns of genetic diversity in North American forest trees.

Region and species	Family	Genetic markers	Number of groups	Pattern	Reference(s)
Southeastern North America					
<i>Abies fraseri</i>	Pinaceae	cpDNA SSRs	1 (3?)	No clear pattern. Possible north-south subdivision along the Appalachian Mountains	Potter 2006
<i>Acer rubrum</i>	Aceraceae	cpDNA sequences	3	Divergence of the central Florida stands. Colonization from northern populations	McLachlan et al. 2005
<i>Chamaecyparis thyoides</i>	Cupressaceae	Allozymes	3	Differentiation between Gulf and Atlantic populations	Mylecraine et al. 2004
<i>Fagus grandifolia</i>	Fagaceae	Allozymes, cpDNA sequences	3	Subdivision between the Gulf and south Atlantic stands and the northern populations up to New England. Colonization from northern populations. Divergence of isolated Mexican stands	Kitamura and Kawano 2001; Rowden et al. 2004; McLachlan et al. 2005
<i>Juglans nigra</i>	Juglandaceae	Nuclear SSRs	1	No clear pattern. Putative recolonization from a single glacial population	Victory et al. 2006
<i>Liquidambar styraciflua</i> complex	Altingiaceae	Allozymes, cpDNA sequences	2 (3?)	Ancient vicariance. Putative differentiation east-west of the Mississippi Valley and east-west of the Appalachians	Wells et al. 1991; Morris et al. 2008
<i>Liriodendron tulipifera</i>	Magnoliaceae	cpDNA restriction sites, allozymes	2	Differentiation between the Gulf and Atlantic populations. North-south subdivision in Atlantic stands	Parks et al. 1994; Sewell et al. 1996
<i>Magnolia fraseri</i> , <i>M. macrophylla</i> , and <i>M. tripetala</i>	Magnoliaceae	Allozymes	1	No clear pattern. Putative recolonization from a single glacial population	Qiu and Parks 1994
<i>Pinus clausa</i>	Pinaceae	Allozymes	2	Differentiation between central Florida and Gulf stands	Parks et al. 1994
<i>Pinus echinata</i>	Pinaceae	Allozymes	1	Putative expansion from a single refuge in central Texas	Schmidtting 2003
<i>Pinus virginiana</i>	Pinaceae	Allozymes	2	Differentiation between populations southeast and northwest of the Appalachians	Parker et al. 1997
<i>Pinus palustris</i>	Pinaceae	Allozymes	2	Possible differentiation east-west of the Mississippi Valley with postglacial expansion from west to east	Schmidtting and Hipkins 1998
<i>Pinus rigida</i>	Pinaceae	Allozymes	1	No clear pattern. Putative recolonization from a single glacial population	Guries and Ledig 1982
<i>Pinus taeda</i>	Pinaceae	Allozymes, nuclear SSRs	2	Differentiation east-west of the Mississippi Valley	Schmidtting et al. 1999; Al-Rabab'ah and Williams 2004
<i>Prunus</i> section <i>Prunocerasus</i>	Rosaceae	cpDNA sequences	7	No clear structure. Putative isolated refuge in central Texas	Shaw and Small 2005
<i>Quercus rubra</i>	Fagaceae	cpDNA restriction sites	1	No clear pattern. Putative recolonization from a single glacial population	Magni et al. 2005
Boreal North America					
<i>Acer saccharum</i>	Aceraceae	Allozymes	2	Differentiation between the Maritime-Quebec and central Ontario stands	Young et al. 1993
<i>Picea glauca</i>	Pinaceae	cpDNA sequences	2	Putative isolated refuge in Alaska	Anderson et al. 2006
<i>Picea mariana</i> – <i>Picea rubens</i> complex	Pinaceae	mtDNA sequences and restriction sites	5	Differentiation among the eastern, central-continental, and western stands and putative northeastern glacial refuge for <i>Picea mariana</i> . Putative mtDNA recombination in the zone of contact between <i>Picea mariana</i> and <i>Picea rubens</i>	Gamache et al. 2003; Jaramillo-Correa et al. 2004; Jaramillo-Correa and Bousquet 2005
<i>Pinus banksiana</i>	Pinaceae	mtDNA STR	4	Differentiation among the eastern and central-continental stands. Putative northeastern glacial refuge. Introgression with <i>Pinus contorta</i> in the west	Godbout et al. 2005
<i>Pinus resinosa</i>	Pinaceae	cpDNA SSRs	2	Differentiation among eastern and central-continental stands. Putative northeastern glacial refuge	Echt et al. 1998; Walter and Epperson 2001

Table 1 (continued).

Region and species	Family	Genetic markers	Number of groups	Pattern	Reference(s)
Pacific Northwest					
<i>Abies procera</i>	Pinaceae	Allozymes	2 (?)	Divergence of coastal isolated stands. North-south differentiation along the Cascades possibly produced by introgression with <i>Abies magnifica</i> in southern populations	Yeh and Hu 2005
<i>Alnus rubra</i>	Betulaceae	cpDNA restriction sites	2	North-south discontinuity among the coastal stands	Soltis et al. 1997
<i>Betula occidentalis</i>	Betulaceae	Allozymes	1	No clear pattern. Putative recolonization from a single glacial population with isolation by distance. Introgression with <i>Betula papyrifera</i> at the north of its range	Williams and Arnold 2001
<i>Chamaecyparis nootkatensis</i>	Cupressaceae	Allozymes	3	North-south discontinuity among the coastal stands	Ritland et al. 2001
<i>Larix lyallii</i>	Pinaceae	Nuclear SSRs	2	Putative east-west differentiation in the northern Rocky Mountains	Khasa et al. 2006
<i>Larix occidentalis</i>	Pinaceae	Nuclear SSRs	2	Putative east-west differentiation in the northern Rocky Mountains	Khasa et al. 2006
<i>Picea engelmannii</i>	Pinaceae	Allozymes	2	North-south differentiation along the Rocky Mountains. Introgression with <i>Picea glauca</i> at the north of its range	Ledig et al. 2006b
<i>Picea sitchensis</i>	Pinaceae	Nuclear SSRs and ESTPs	1 (?)	Rapid decline of genetic diversity towards the species margins. Isolation by distance	Gapare et al. 2005; Mimura and Aitken 2007
<i>Pinus albicaulis</i>	Pinaceae	mtDNA, cpDNA SSRs	3-6	Discontinuity between the Sierran, Cascades, and Rocky Mountains populations. North-south differentiation within the last two regions	Richardson et al. 2002
<i>Pinus contorta</i> complex	Pinaceae	Allozymes, RAPDs, mtDNA, cpDNA SSRs	5	No correlation between genetic structure and differences among subspecies. Discontinuity between coastal and montane stands. North-south differentiation along the Pacific Coast and the Rocky Mountains. Quick postglacial expansion with isolation by distance. Putative cpDNA recombination in the zone of contact with <i>Pinus banksiana</i>	Wheeler and Guries 1982; Marshall et al. 2001, 2002; Fazekas and Yeh 2006; Godbout et al. 2008
<i>Pinus flexilis</i>	Pinaceae	Allozymes, mtDNA	7	Discontinuity between Sierran and Rocky Mountains populations. Modern stands from the Rockies derived from several isolated refugia. Restricted Holocene gene flow	Mitton et al. 2000; Jørgensen et al. 2002
<i>Pinus monticola</i>	Pinaceae	Allozymes	2	North-south discontinuity along the Cascades	Steinhoff et al. 1983
<i>Pinus ponderosa</i> complex	Pinaceae	mtDNA	4	Sierran-Cascades - Rocky Mountains discontinuity. Holocene introgression between varieties. North-south subdivision along the Rocky Mountains. Ancient vicariance	Latta and Mitton 1999; Johansen and Latta 2003
<i>Populus trichocarpa</i>	Salicaceae	Allozymes	2 (?)	Possible north-south subdivision along the Cascades	Weber and Stettler 1981
<i>Pseudotsuga menziesii</i>	Pinaceae	Allozymes, cpDNA SSRs	4	Divergence between coastal and interior varieties. North-south subdivision in interior variety. Divergence of Mexican stands	Li and Adams 1989; Aagard et al. 1998; Viard et al. 2001
<i>Salix melanopsis</i> complex	Salicaceae	cpDNA sequences	4	Cascades - Rocky Mountains discontinuity. North-south differentiation along the Rocky Mountains. Ancient vicariance	Carstens et al. 2005; Brunfeldt et al. 2007
<i>Thuja plicata</i>	Cupressaceae	RFLPs, nuclear SSRs	3	Subdivision among northern-coastal, central and southern-interior stands. Putative recolonization from a single glacial population.	Glaubitz et al. 2000; O'Connell et al. 2008

Table 1 (continued).

Region and species	Family	Genetic markers	Number of groups	Pattern	Reference(s)
California					
<i>Abies bracteata</i>	Pinaceae	Allozymes	1	No apparent differentiation among isolated stands.	Ledig et al. 2006a
<i>Abies magnifica</i>	Pinaceae	cpDNA sequences	2	Discontinuity between the Sierra Nevada and the Klamath – southern Cascades stands. Putative introgression from <i>Abies procera</i>	Oline 2008
<i>Chamaecyparis lawsoniana</i>	Cupressaceae	Allozymes	2 (?)	Putative differentiation between inland and coastal stands	Millar and Marshall 1991
<i>Picea breweriana</i>	Pinaceae	Allozymes	1	No clear pattern. Population divergence mainly modeled by genetic drift	Ledig et al. 2005
<i>Pinus attenuata</i> , <i>P. radiata</i> , <i>P. muricata</i>	Pinaceae	Allozymes, RAPDs, cpDNA and mtDNA restriction sites	—	Ancient vicariance among species with putative south–north migration. North–south differentiation along coastal populations observed with cytoplasmic markers for <i>Pinus muricata</i>	Hong et al. 1993; Wu et al. 1998, 1999
<i>Pinus balfouriana</i>	Pinaceae	Allozymes, cpDNA, mtDNA and nuclear DNA sequences	1	No clear genetic structure reflecting the geographic differentiation between varieties or among populations of the same variety.	Oline et al. 2000; Eckert et al. 2008
<i>Pinus coulteri</i>	Pinaceae	Allozymes	1	Ancient vicariance Migration from a single southern refugia with clinal loss of diversity. Putative introgression with <i>Pinus jeffreyi</i> in sympatric populations	Ledig 2000
<i>Pinus jeffreyi</i>	Pinaceae	Allozymes	2	Discontinuity between Sierran and Coastal ranges populations	Furnier and Adams 1986
<i>Pinus lambertiana</i>	Pinaceae	cpDNA sequences	2	North–south disjunction along the Sierra Nevada. Putative introgression from <i>Pinus albicaulis</i> in the northern part of the distribution	Liston et al. 2007
<i>Pinus sabiniana</i>	Pinaceae	Allozymes	1	Holocene collapse and fragmentation. No clear geographic pattern	Ledig 1999
<i>Quercus agrifolia</i> , <i>Q. kelloggii</i> , <i>Q. parvula</i> , and <i>Q. wislizeni</i>	Fagaceae	AFLPs	—	Ancient vicariance and strong introgression by pollen among species. Coastal and Sierran divergence among stands of <i>Quercus wislizeni</i>	Dodd and Kashani 2003; Dodd and Afzal-Rafii 2004
<i>Quercus lobata</i>	Fagaceae	Nuclear and cpDNA SSRs	—	Ancient vicariance. Patchy distribution of genetic diversity with east–west partition. Limited introgression with other <i>Quercus</i> species	Grivet et al. 2006, 2008
<i>Quercus</i> section <i>Protobalanus</i>	Fagaceae	cpDNA sequences	2	Ancient vicariance. North–south disjunction of major chlorotypes among species along the Sierra Nevada. Putative Quaternary introgression	Manos et al. 1999
Mexico and northern Central America					
<i>Abies flinckii</i> , <i>A. guatemalensis</i> , <i>A. hickeli</i> , <i>A. religiosa</i>	Pinaceae	Allozymes, mtDNA and cpDNA SSRs	9	Divergence of isolated stands along the Transverse Volcanic Belt and at the limits of the species' distribution. Differentiation mainly modeled by genetic drift after Holocene collapse	Aguirre-Planter et al. 2000; Jaramillo-Correa et al. 2008
<i>Picea chihuahuana</i>	Pinaceae	Allozymes, cpDNA SSRs, mtDNA sequences	2	North–south differentiation along the Sierra Madre Occidental observed with mtDNA markers. Population divergence mainly modeled by genetic drift	Ledig et al. 1997; Jaramillo-Correa et al. 2006
<i>Picea martinezii</i>	Pinaceae	Allozymes, cpDNA, mtDNA sequences	1	Almost no differentiation among stands	Ledig et al. 2000a; Jaramillo-Correa et al. 2006
<i>Picea mexicana</i>	Pinaceae	Allozymes, cpDNA, mtDNA sequences	3	Virtually all populations differ from each other. Divergence mainly modeled by genetic drift after Holocene collapse	Ledig et al. 2002; Jaramillo-Correa et al., unpublished

Table 1 (concluded).

Region and species	Family	Genetic markers	Number of groups	Pattern	Reference(s)
<i>Pinus chiapensis</i>	Pinaceae	RAPDs, mtDNA restriction sites	11	Virtually all populations differ from each other. Divergence mainly modeled by genetic drift after Holocene collapse	Newton et al. 2002
<i>Pinus greggii</i>	Pinaceae	Allozymes	3	Divergence of isolated stands along the Sierra Madre Oriental mainly modeled by genetic drift during the Holocene	Parraguirre-Lezama et al. 2002
<i>Pinus montezumae</i> – <i>Pinus hartwegii</i> complex	Pinaceae	cpDNA restriction sites	4	Divergence of isolated stands along the Transverse Volcanic Belt and the Sierra Madre Oriental. Introgressive hybridization between the two taxa	Matos and Schaal 2000
<i>Pinus montezumae</i> – <i>Pinus pseudostrobus</i> complex	Pinaceae	cpDNA SSRs	3	Ancient vicariance with lineage sorting. Introgressive hybridization between the two taxa. Isolation by distance among stands of the same species	Delgado et al. 2007
<i>Pinus nelsonii</i>	Pinaceae	cpDNA SSRs	1	Low population differentiation probably associated with recent divergence. Isolation by distance among stands	Cuenca et al. 2003
<i>Pinus oocarpa</i>	Pinaceae	Allozymes, RAPDs, AFLPs	2 (?)	Low divergence among populations. Putative introgression with <i>Pinus patula</i> subsp. <i>tecunumanii</i>	Matheson et al. 1989; Díaz et al. 2001
<i>Pinus pinceana</i>	Pinaceae	Allozymes	3	North-south divergence along the Sierra Madre Oriental probably related to Holocene population collapse	Ledig et al. 2001
<i>Pinus rzedowskii</i>	Pinaceae	Allozymes	2	Divergence of isolated stands along the Transverse Volcanic Belt mainly modeled by genetic drift during the Holocene	Delgado et al. 1999
<i>Pinus strobiformis</i> – <i>Pinus ayacahuite</i> complex	Pinaceae	cpDNA SSRs	5	Divergence between the Sierra Madre Occidental and Sierra Madre Oriental stands. Differentiation of some isolated populations. Introgressive hybridization and (or) shared ancestral polymorphism between both taxa	Moreno-Letelier and Piñero 2008
<i>Quercus affinis</i> – <i>Quercus laurina</i> complex	Fagaceae	RAPDs, cpDNA restriction sites	2	Ancient vicariance with strong Holocene introgression	González-Rodríguez et al. 2004
<i>Quercus crassifolia</i> – <i>Quercus crassipes</i> complex	Fagaceae	RAPDs	2	Ancient vicariance with strong Holocene introgression	Tovar-Sánchez and Oyama 2004

revealed a genetic discontinuity between the Gulf and South Atlantic populations and the northern stands up to New England (Kitamura and Kawano 2001).

In the southern part of the central plains, the most commonly observed genetic break occurs between the two sides of the Mississippi Valley (Fig. 1C) (Soltis et al. 2006). In forest trees, this discontinuity was first suspected from the phenotypic variation of *Pinus taeda* and *Liquidambar styraciflua* (Wells et al. 1991) and was then confirmed for the former taxon with the analysis of nuclear allozymes (Schmidtling et al. 1999) and microsatellites (Al-Rabab'ah and Williams 2004). Similar studies in *Pinus palustris* (Schmidtling and Hipkins 1998), *Pinus echinata* (Schmidtling 2003), and some species of *Prunus* (Shaw and Small 2005) further revealed that the populations from central Texas were distinct from the neighboring stands of the Mississippi Valley. These results were interpreted as evidence of a distinct refuge in central Texas during the last glacial maximum.

Whether the Appalachian Mountains acted as a refugial place or as a vicariance factor in the past is still a source of debate. Phylogeographic studies have indeed suggested that this range could have prompted genetic differentiation in some species such as *Chamaecyparis thyoides* (Fig. 1B) (Mylecraine et al. 2004), *Liquidambar styraciflua* (Morris et al. 2008), and *Pinus banksiana* (Godbout et al. 2005) (see below). However, in other taxa such as *Quercus rubra* (Magni et al. 2005), *Acer rubrum* (McLachlan et al. 2005), *Juglans nigra* (Victory et al. 2006), and three different species of *Magnolia* (Qiu and Parks 1994), the distribution of nuclear and cpDNA genetic diversity did not reflect any particular genetic discontinuity associated with this mountain range. On the other hand, the distribution of allozyme diversity in *Tsuga canadensis* points to the southeastern Appalachians as a putative glacial refuge for this species (Potter et al. 2007).

Boreal North America

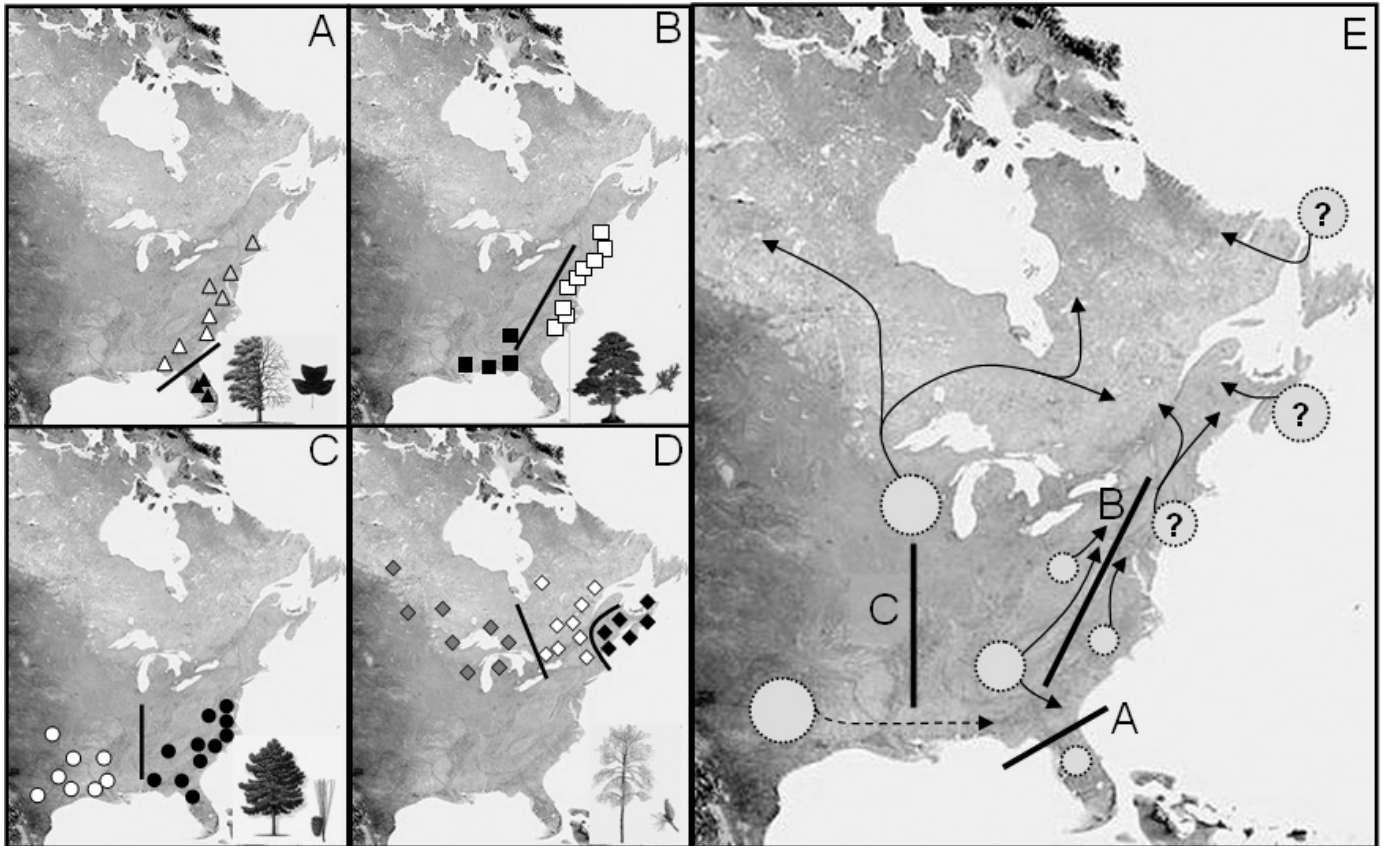
The most common phylogeographic pattern observed in the northern part of the continent is the division between the different eastern lineages restricted to eastern Quebec and the Maritime Provinces of Canada and the one or few widespread central lineages distributed from the Great Lakes region to the eastern slopes of the Rocky Mountains. This genetic discontinuity is particularly visible in broad-range species such as *Abies balsamea*, *Acer saccharum*, *Picea mariana*, *Pinus banksiana*, and *Pinus resinosa* among others (Table 1) (Young et al. 1993; Echt et al. 1998; Clark et al. 2000; Walter and Epperson 2001; Jaramillo-Correa et al. 2004; Godbout et al. 2005). For example, in *Pinus banksiana*, all of the populations located between central Ontario and eastern Alberta shared the very same mitotypes, while there were two different genetic lineages restricted to the St. Lawrence and Ottawa River valleys and to southern Quebec, New Brunswick, and Nova Scotia (Fig. 1D) (Godbout et al. 2005). This pattern suggests that all of the central and western stands might have been derived from the same glacial population, while the eastern stands originated from different isolated eastern refugia (Godbout et al. 2005). Indeed, previous palynological and paleoecological surveys have suggested that, during the interglacials, the glacial popula-

tions located south of the Great Lakes region expanded rapidly to the northwest, colonizing a large part of central and western Canada (e.g., Matthews et al. 1989; McLeod and MacDonald 1997), while the eastern populations spread north following the Atlantic coastline at a slower pace (Fig. 1E) (Matthews et al. 1989).

Additional paleobotanical and geological studies have also shown that some northern regions of the continent were never or were only partially covered by ice during some of the glacial maxima (e.g., Edwards et al. 2000; Goetcheus and Birks 2001). However, whether these unglaciated lands were able to sustain viable populations that contributed to the postglacial colonization of North America is still a source of debate. Many of these paleobotanical studies point to Beringia as the most plausible northern refugial region. This zone was probably inhabited by tundra shrub species of the genera *Alnus*, *Betula*, *Salix*, and *Rhododendron* during the glacial maxima (e.g., Edwards et al. 2000). However, some genetics studies, including the analysis of cpDNA variation in *Picea glauca* (Anderson et al. 2006), have pointed out that boreal forest tree species might have also been present in this region during the last glacial maximum. Similar evidence recently found for *Picea mariana* (S. Gerardi et al., unpublished results) tends to support this hypothesis.

Further phylogeographic studies have also suggested other potential northern refuges in regions that were apparently ice-free during some of the glacial cycles such as the Canadian High Arctic, the corridor between the Laurentide and Cordilleran ice sheets, and different emerged banks on the northeastern coast between Newfoundland and New England (e.g., Ehlers and Gibbard 2004). For example, the distribution of chlorotypes in *Saxifraga oppositifolia* (Abbot et al. 2000) and *Dryas integrifolia* (Tremblay and Schoen 1999), two annual plants, pointed to the existence of a few northern refugia in the Canadian High Arctic. However, the analysis of mtDNA variation in high-latitude *Picea mariana* populations suggested that these regions did not serve as glacial refugia for this and perhaps other forest tree taxa (Gamache et al. 2003). The existence of glacial refuges in the ice-free corridor between the Laurentide and Cordilleran ice sheets is still controversial, as reflected by the different genetic surveys conducted in various species (see Godbout et al. 2008 and references therein), while the presence of boreal tree species in different places of the emerged coasts between Newfoundland and New Jersey seems better supported by several phylogeographic studies (e.g., Walter and Epperson 2001; Jaramillo-Correa et al. 2004; Godbout et al. 2005). Different geological surveys have further shown that various portions of the Scotian Shelf were emerged and ice-free during the last glaciation (e.g., King 1996; Shaw and Gareau 2002) and point to the possibility that this series of "ice-free islands" might have played an important role in the postglacial recolonization of this part of the continent (e.g., Keddy and Wisheu 1989; Klimaszewski et al. 2006). However, the difficulty in finding fossils, pollen, or macroremnants in this and other potential northern refugial places hampers their study for the time being. The integration of new laboratory detection and analytical techniques might prompt the confirmation of their existence and their integration into simulation analyses to understand their role in the

Fig. 1. (A–D) Examples of genetic discontinuities observed in forest trees from eastern North America including the boreal region. Populations with similar symbols belong to the same genetic lineage as revealed by different clustering analyses. (A) Discontinuity between the populations of central Florida and the Gulf and East coasts as revealed by a principal component analysis on allozyme and cpDNA variation of *Liriodendron tulipifera* (redrawn from Parks et al. 1994 and Sewell et al. 1996). (B) Discontinuity between the populations on both slopes of the Appalachian Mountains as shown by an UPGMA clustering of Nei's unbiased distances on allozyme data of *Chamaecyparis thuyoides* (redrawn from Mylecraine et al. 2004). (C) Discontinuity between the populations on both sides of the Mississippi Valley as observed with a canonical discriminant analysis on allozyme frequencies of *Pinus taeda* (redrawn from Schmidting et al. 1999). (D) Discontinuities between the central and eastern boreal populations as observed with a Montmonier analyses on mtDNA variation of *Pinus banksiana* (redrawn from Godbout et al. 2005). (E) Summary of the most common genetic discontinuities (solid lines) and location of the putative glacial populations (circles) and postglacial colonization routes (arrows) reported on palynological and phylogeographic studies of forest trees of the region.



recolonization of North America (see Provan and Bennett 2008 for a further discussion on this topic).

Pacific Northwest

This region spans from the coasts of southern Alaska to those in northern California and east to the Rocky Mountains in Alberta, central Idaho, and western Montana (Brunsfeld et al. 2001). It encompasses mostly mesic and temperate coniferous forests and is divided longitudinally by two western coastal ranges (the Cascades and Coast Range) and by the Rocky Mountains. These mountain ranges apparently define most of the genetic discontinuities observed in this region (Fig. 2) (Brunsfeld et al. 2001, 2007; Carstens et al. 2005).

One of the first comparative phylogeographic studies made in the Pacific Northwest (Soltis et al. 1997) contrasted the distribution of cpDNA diversity in six coastal plant species including *Alnus rubra* and *Picea sitchensis*, two forest tree taxa. For most of these plants, the cpDNA lineages were divided into two major groups: a northern group dis-

tributed from southern Alaska to northern Oregon and a southern lineage spanning from central Oregon to northern California (Streng 1994; Soltis et al. 1997). This north-south subdivision among coastal stands or among populations located in the same mountain range was also observed with nuclear, cpDNA, and mtDNA markers in *Thuja plicata* (Glaubitz et al. 2000; O'Connell et al. 2008), *Chamaecyparis nootkatensis* (Ritland et al. 2001), *Abies procera* (Yeh and Hu 2005), *Salix melanopsis* (Brunsfeld et al. 2007), *Pinus contorta* (Wheeler and Guries 1982; Fazekas and Yeh 2006; Godbout et al. 2008), and other animal and plant species (reviewed in Brunsfeld et al. 2001). This discontinuity has often been attributed to northwards long-distance colonization events or to the existence of genetically distinct glacial populations in the Queen Charlotte (Haida Gwaii) islands (e.g., Glaubitz et al. 2000; Godbout et al. 2008).

A second genetic break observed in this part of the continent occurs between the coastal and interior regions. In forest trees, this discontinuity was first detected from the morphological races of *Pinus contorta*, *Pinus ponderosa*,

and *Pseudotsuga menziesii* (e.g., Wells 1964; Burns and Honkala 1990) and was later observed in the distribution of their genetic lineages (Figs. 2A and 2B) (Li and Adams 1989; Latta and Mitton 1999; Viard et al. 2001; Marshall et al. 2002; Johansen and Latta 2003; Godbout et al. 2008). Additional forest tree taxa showing this phylogeographic break include *Pinus albicaulis* (Fig. 2C) (Richardson et al. 2002), *Pinus flexilis* (Latta and Mitton 1997; Mitton et al. 2000), *Thuja plicata* (O'Connell et al. 2008), and *Salix melanopsis* (Fig. 2D) (Carstens et al. 2005; Brunsfeld et al. 2007). This break has indeed been observed in more than 150 different plant, animal, and fungal species and has been associated with three different phylogeographic hypotheses (reviewed by Carstens et al. 2005): (i) ancient vicariance, which implies the persistence of both coastal and inland glacial refuges during most of the Pleistocene (e.g., Brunsfeld et al. 2007; Godbout et al. 2008), (ii) inland dispersal from a northern refuge, which implies that the Rocky Mountains were colonized after the Pleistocene from refugia located in the northern Cascades following routes along the margins of the retreating glaciers (e.g., Richardson et al. 2002), and (iii) inland dispersal from southern refuges, which suggests the colonization of the Rocky Mountains from refugia located in the southern parts of the Cascades (e.g., Strenge 1994).

Californian Floristic Province

California and the northern part of the Baja California Peninsula encompass one of the most geographically and geologically complex landscapes on earth (Myers et al. 2000). It comprises at least three main mountain ranges, which have played a mayor role in generating isolation and speciation during the last 5 million years (Calsbeek et al. 2003; Lapointe and Rissler 2005). The local climatic regimes have apparently been key in shaping the distribution of genetic diversity by promoting adaptation of specific genetic lineages to each particular microclimate (e.g., Dodd and Afzal-Rafii 2004; Lapointe and Rissler 2005; Grivet et al. 2006, 2008).

Although some exceptions exist (e.g., Ledig 1999; Ledig et al. 2006a), populations from this region can usually be divided into three geographically structured groups: a northern one that sometimes spreads north into the Pacific Northwest, a restricted eastern Sierran, and a southern group that is seldom dispersed south into northwestern Mexico (Table 1; Figs. 2E and 2F) (Furnier and Adams 1986; Manos et al. 1999; Calsbeek et al. 2003; Shaw and Small 2005; Liston et al. 2007 among others). As reported for Californian oaks (Manos et al. 1999; Dodd and Kashani 2003; Dodd and Afzal-Rafii 2004; Grivet et al. 2006, 2008) and pines (Hong et al. 1993; Strauss et al. 1993; Wu et al. 1998, 1999; Liston et al. 2007), differentiation among these three lineages usually dates back to the Tertiary, while their relocation at the local scale, via migration, hybridization, and introgression, should have been prompted by the Pleistocene glacial cycles (also see Ledig 2000; Lapointe and Rissler 2005; Grivet et al. 2006).

Mexico

When considering the geographical and geological complexity of Mexico, it would seem inappropriate to treat this entire country as one region. However, the limited num-

ber of genetic, biogeographic, and palynological studies conducted in Mexican forest trees hampers, for the time being, any attempt for a more detailed review at a finer geographical scale.

Mexico encompasses an extremely diverse and heterogeneous mixture of temperate, subtropical, and tropical forest tree species, which are distributed from the northern arid lowlands and the humid montane environments to the southern dry and wet tropical forests (Perry et al. 2000; Luna et al. 2001). Only a few of the modern Mexican tree genera are endemic, while most of them migrated from northern North America during the Tertiary or from northern South America during the Quaternary (e.g., Graham 1973, 1995). Different palynological studies have shown that most of these taxa, especially those from the montane regions, have undergone recurrent latitudinal and altitudinal migration cycles since their establishment (Graham 1973). Such cycles have produced complex evolutionary and biogeographic patterns by promoting the expansion, fragmentation, and divergence of populations and the hybridization and introgression of closely related species (Table 1) (e.g., Aguirre-Planter et al. 2000; Matos and Schaal 2000; González-Rodríguez et al. 2004; Jaramillo-Correa et al. 2006; Delgado et al. 2007 among others).

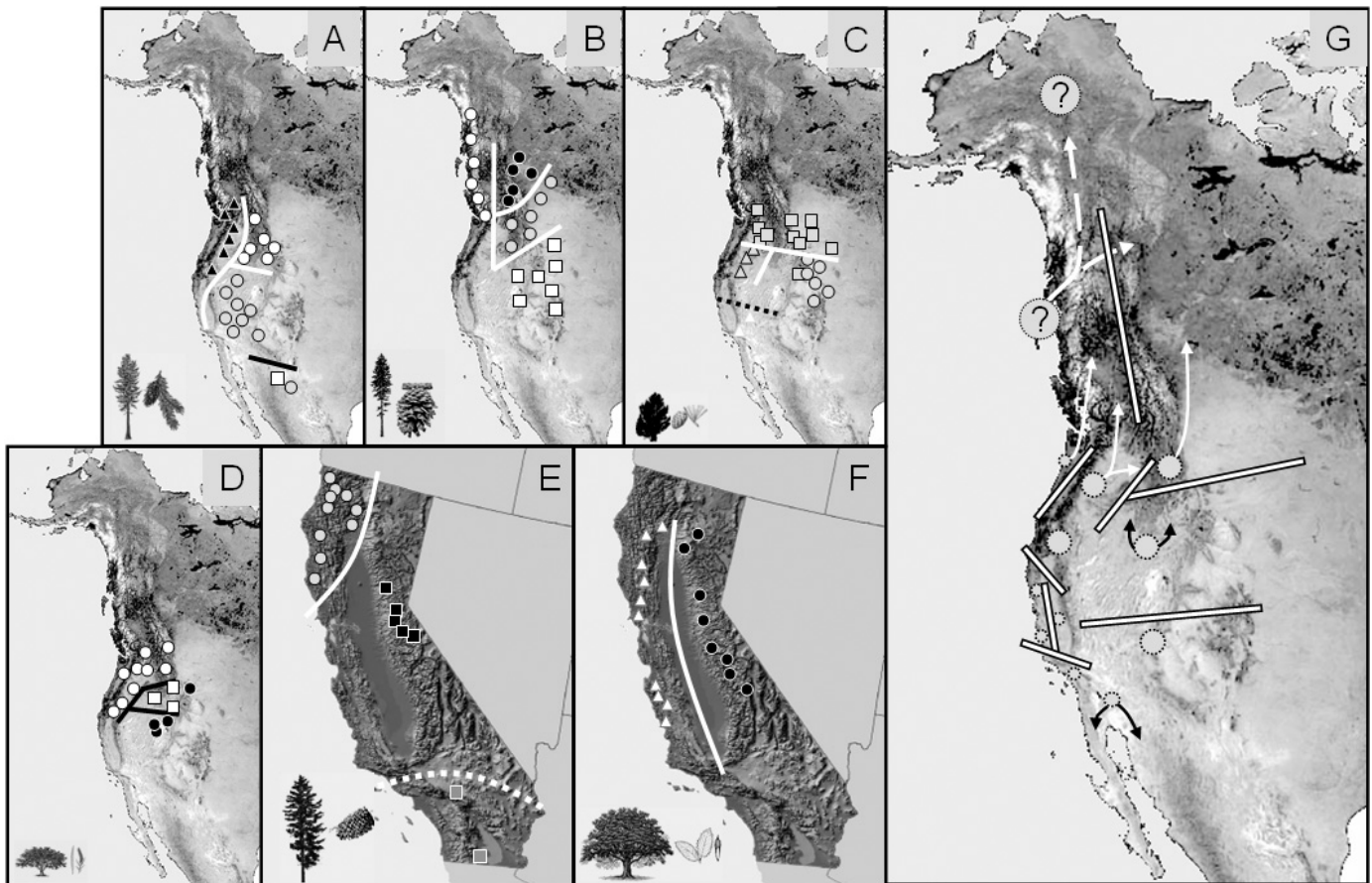
Although some consistent subdivisions have been observed at the biogeographic and systematic levels (reviewed in Morrone 2005), it has usually been difficult to find congruent genetic discontinuities among co-distributed Mexican taxa (Table 1; Fig. 3). Indeed, it seems that each Mexican species bears a different phylogeographic signature depending on its particular life history traits and on the strength of the selective and stochastic forces that it has coped with. For example, the distribution of chlorotypes in closely related *Quercus* species was not related to either the geographic location or the taxonomic status of the populations surveyed (González-Rodríguez et al. 2004). A similar pattern was also observed with nuclear and mtDNA markers in *Pinus chiapensis* (Newton et al. 2002), with cpDNA markers in *Pinus nelsonii* (Cuenca et al. 2003), and with nuclear and mtDNA markers in four *Abies* species from southern Mexico and Guatemala (Aguirre-Planter et al. 2000; Jaramillo-Correa et al. 2008). However, the distribution of chlorotypes in these same *Abies* taxa, in *Pinus strobiformes*, as well as in the *Pinus montezumae* – *Pinus hartwegii* and *Pinus montezumae* – *Pinus pseudostrobus* complexes (Matos and Schaal 2000; Delgado et al. 2007; Moreno-Letelier and Piñero 2008) appeared to be geographically structured. A similar pattern was observed with allozymes in *Pinus pinceana* (Ledig et al. 2001) and with mtDNA markers in *Picea chihuahuana* (Jaramillo-Correa et al. 2006). Nevertheless, only a few of these reported genetic discontinuities appear to coincide with the currently recognized Mexican biogeographic regions (Fig. 3).

Evolutionary trends depicted in phylogeographic studies of North American forest trees

Isolation, genetic pauperization, and extinction

Population isolation is usually accompanied by the action of stochastic forces such as genetic drift and inbreeding,

Fig. 2. (A–F) Examples of common genetic discontinuities observed in forest trees from western North America including California and the boreal regions. Populations with similar symbols belong to the same genetic lineage as revealed by different clustering analyses. (A) Discontinuity between the coastal and mountain populations and between the northern and southern stands of the same mountain range as observed with a UPGMA cluster analysis of Nei's unbiased distances of allozyme data from *Pseudotsuga menziesii* (redrawn from Li and Adams 1989) (B) Similar discontinuity detected with a UPGMA clustering of $\delta\mu^2$ distances of mtDNA variation of *Pinus contorta* (redrawn from Godbout et al. 2008). (C) Discontinuity between the populations of different mountain ranges as shown by the distribution of mitotypes in *Pinus albicaulis* populations from the Sierra Nevada, the Cascades, and the Rocky Mountains (redrawn from Richardson et al. 2002). (D) Similar discontinuity observed with the distribution of chlorotypes across stands of *Salix melanopsis* from the Cascades and the Rocky Mountains (redrawn from Brunsfeld et al. 2007). (E) Discontinuity between the coastal and montane populations in California as reported from a UPGMA cluster analysis of Nei's unbiased distances of allozyme data from *Pinus jeffreyi* (redrawn from Furnier and Adams 1986). (F) East–west discontinuity detected with a principal coordinates analysis on AFLP data from *Quercus wislizeni* (redrawn from Dodd and Kashani 2003). (G) Summary of the most common genetic discontinuities (solid lines) and location of the putative glacial populations (circles) and postglacial colonization routes (arrows) reported on palynological and phylogeographic studies of forest trees of the region.



which usually leads to a more or less rapid loss of genetic diversity. Most of the North American forest trees exhibiting low levels of genetic diversity are currently secluded in small isolated stands or are believed to have been isolated in small glacial refugia in the past. Generally, genetic pauperization is more easily detected with organelle genomes due to their haploid nature and smaller effective population size (Birky et al. 1989). However, in some extreme cases, such as in *Pinus resinosa*, *Picea chihuahuana*, *Picea martinezii*, and *Larix lyallii*, genetic pauperization is also observed at nuclear loci (e.g., Fowler and Morris 1977; DeVerno and Mosseler 1997; Ledig et al. 1997, 2000a; Khasa et al. 2006). Most of these reported cases occur in montane, coastal, or insular species, especially along the

mountain ranges and in the southern parts of the continent, where most of the endemic and relict taxa are found. Interestingly, many of these taxa are conifers (see Fig. 4 and references therein).

Palynological studies have suggested that some of these genetically depauperate species had broader ranges in the past and then collapsed into small and scattered island-like stands due to various ecological reasons during the Pleistocene and Holocene climate changes (e.g., Jackson et al. 1996). However, although some authors have suggested that most of these montane taxa might disappear with rapid global warming (e.g., Thomas et al. 2004), these remnant populations, if conserved, may also serve as interglacial refugia from which species could reexpand if the climatic

trends were to reverse (Ledig et al. 2000a; Hampe and Petit 2005). Indeed, there are some well-documented examples of species that have developed selfing-tolerant systems that have allowed them to survive genetic pauperization and even have favored their ability to colonize new habitats (e.g., Fowler 1965; Ledig et al. 2000a; also see Petit and Hampe 2006 for a review). This is the case for *Pinus resinosa*, a broad-range boreal-temperate conifer, which was able to colonize a large part of the boreal and temperate forests in northeastern North America after losing most of its genetic diversity during the Pleistocene (Fowler and Morris 1977; DeVerno and Mosseler 1997; Walter and Epperson 2001). Other evolutionary and adaptive traits, such as the capacity to change the allocation of resources between survival, growth, and reproduction, might indeed render forest trees particularly resilient to extinction (see Hampe and Petit 2005 for a review). One such extreme case is *Picea mariana* at the northern limit of its distribution, where it can quickly adapt to extreme climatic conditions by shifting resource allocation to survival under the form of *krummholz* clones maintained by layering over centuries and even millennia (Laberge et al. 2000). However, there are many reported cases of forest tree extinctions associated with times of high climate variability (e.g., Jackson and Weng 1999; Petit et al. 2008) and more examples might arise in the future, if adequate conservation strategies are not rapidly implemented.

Speciation and formation of regional races and varieties

The potential role of major historical events, such as the Pleistocene glaciations, in promoting speciation has been a constant source of debate in the population and evolutionary genetics community (e.g., Klicka and Zink 1997; Avise and Walker 1998). Most of this debate revolves around the questions of time and distance: were the periods of population isolation induced by these historical events long enough to allow divergence and speciation? And were the distances among glacial or interglacial refugia large enough to bring these populations to complete isolation from each other?

A number of studies addressing these issues have been conducted, and most of the proposed examples of allopatric speciation deal with short-generation taxa such as annual plants (reviewed by Gottlieb 2004), which seem to have higher diversification rates than long-generation species such as forest trees (e.g., Verdú 2002; Petit and Hampe 2006). Indeed, proposed cases of new forest tree taxa arising after a period of isolation (i.e., a glacial cycle) are rather scarce (e.g., Purdy and Bayer 1995; Perron et al. 2000). Perhaps one of the few well-documented cases at the genetic level for a forest tree species is that of red spruce (*Picea rubens*), which could represent a more or less ancient derivative of black spruce (*Picea mariana*), as a by-product of genetic isolation and allopatric speciation (Perron et al. 2000). However, although the adaptive nature of this divergence has yet to be demonstrated, these two taxa fit the genetic and ecological expectations for a parental-derivative species pair (Perron et al. 2000; Gottlieb 2004).

Additional examples suggesting early stages of speciation in forest trees include morphological races and varieties in *Fagus grandifolia*, *Pseudotsuga menziesii*, and *Pinus ponderosa* among others. These differentiations have been confirmed by the distribution of genetic variation across the

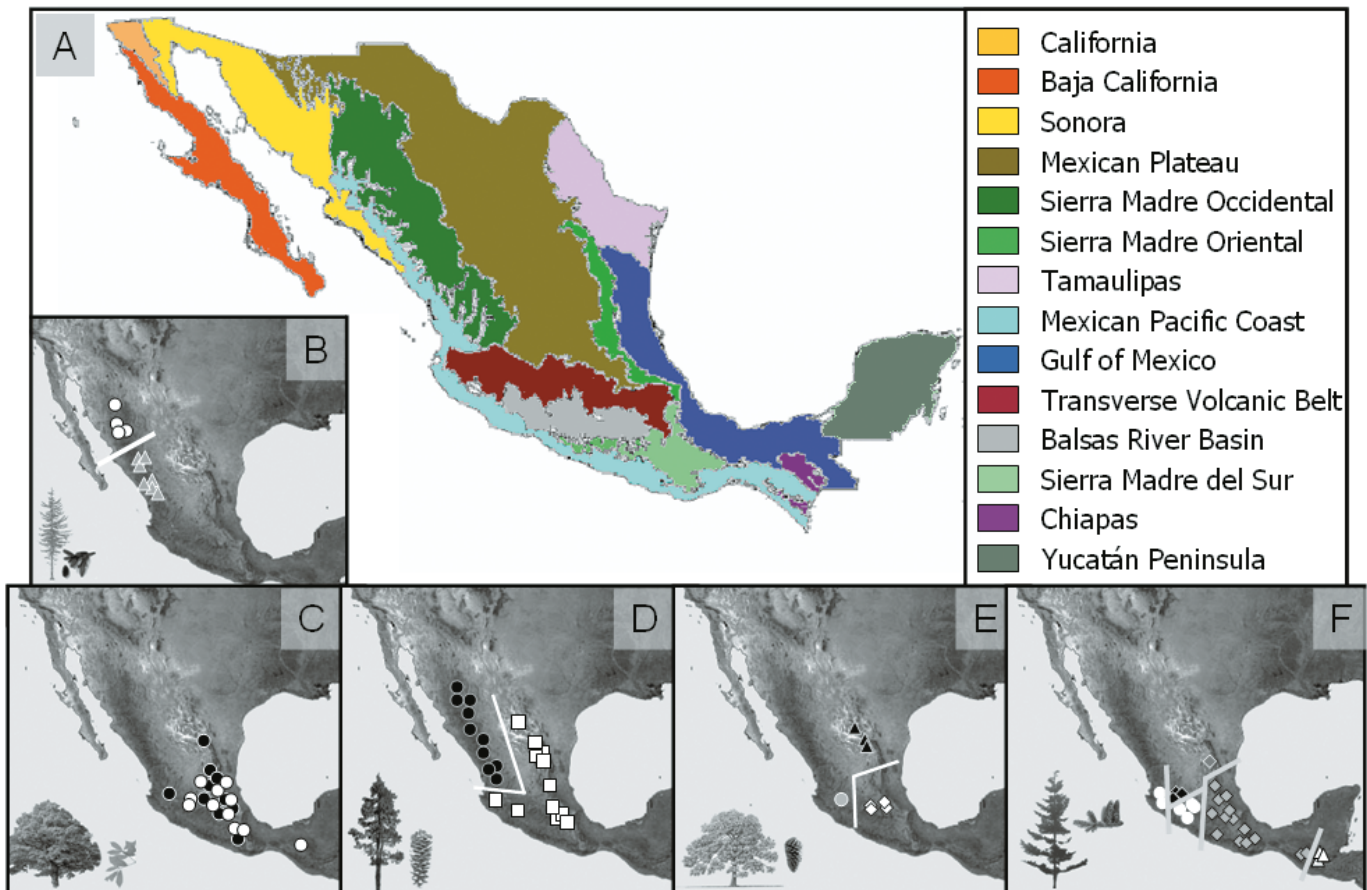
diverging infraspecific taxa (e.g., Li and Adams 1989; Latta and Mitton 1999; Kitamura and Kawano 2001; Rowden et al. 2004). Further phylogeographic studies in other closely related North American forest trees could provide new insights into the role of different historical events in generating divergence and speciation. Particular attention should be paid to tropical and subtropical environments where the number of species is higher and the number of evolutionary studies is still limited.

Secondary contact and suture zones

The term “secondary contact zone” refers to the location where different genetic lineages from the same or from closely related species come into contact after their primary divergence (e.g., Hewitt 1996). Most of the secondary contact zones described for North American forest trees are thought to be the result of range expansions that followed the glacial isolation of diverging lineages. For example, in the northern parts of the continent, the contact zones reported between the two varieties of *Pinus ponderosa*, between the closely related *Picea mariana* and *Picea rubens*, or among different genetic lineages in *Pinus banksiana*, *Pinus taeda*, or *Liriodendron tulipifera* have likely arisen following the last postglacial expansion (e.g., Parks et al. 1994; Perron and Bousquet 1997; Schmidting et al. 1999; Johansen and Latta 2003; Godbout et al. 2005). On the other hand, more ancient zones of contact, such as those occurring in Mexico or California among different species of the genus *Quercus* or *Pinus*, are probably the result of recurrent expansions of the montane forests since their Tertiary or Quaternary speciation (e.g., Wu et al. 1998, 1999; Matos and Schaal 2000; Dodd and Kashani 2003; González-Rodríguez et al. 2004).

The term “suture zone” was first introduced in evolutionary biology by Remington (1968) to describe the regions where multiple zones of hybridization tend to be clustered. However, although some of the regions initially proposed by Remington (1968) have been contested, the initial suture zone concept and its putative evolutionary impacts remain intact (e.g., Hewitt 1996). More recent GIS analyses of superimposed North American species ranges, hybrid zones, and genetic discontinuities (Swenson and Howard 2004, 2005) revealed five main suture zones for forest trees: the Rocky Mountains around the border between the United States and Canada, the borders between Oregon and California, the borders between Illinois and Kentucky, the central and southeastern region of Alabama, and the Sierra Madre Occidental in northwestern Mexico. Additional suture zones including the Great Lakes region and the southern parts of the St. Lawrence Valley in eastern Canada, the Appalachian Mountains and the Arizona–Nevada border in the United States, and the Transverse Volcanic Belt in central Mexico were observed when the ranges of avian and mammalian taxa were also considered (Swenson and Howard 2005). The most important pattern that emerges from the distribution of these zones is their association with mountain ranges, their location between putative glacial refugia, and their colocalization with the southern edges of glacial ice sheets (Swenson and Howard 2005), a picture that resembles the one previously depicted for European species (e.g., Hewitt 1996; Taberlet et al. 1998; Lascoux et al. 2004).

Fig. 3. (A) Mexican biogeographic provinces (modified from Morrone 2005) and (B–F) some of the phylogeographic patterns reported for Mexican forest trees. (B) Discontinuity between northern and southern stands along the Sierra Madre Occidental as observed from the distribution of mitotypes in *Picea chihuahuana* (redrawn from Jaramillo-Correa et al. 2006). (C) Absence of geographic structure in the distribution of chlorotypes in the *Quercus affinis* – *Quercus laurina* complex (redrawn from González-Rodríguez et al. 2004). (D) Discontinuity between the populations of the Sierra Madre Occidental and the Transverse Volcanic Belt and Sierra Madre Oriental as deduced from the distribution of chlorotypes in *Pinus strobiformis* (redrawn from Moreno-Letelier and Piñero 2008). (E) Phylogeographic break between populations in the eastern and western parts of the Transverse Volcanic Belt and the Sierra Madre Oriental as detected with the distribution of chlorotypes in the *Pinus montezumae* – *Pinus hartwegii* complex (redrawn from Matos and Schaal 2000). (F) Differentiation of populations along the Transverse Volcanic Belt and the limits of the distribution of four closely related *Abies* species as detected with a Montmonier analysis on their cpDNA variation (redrawn from Jaramillo-Correa et al. 2008).

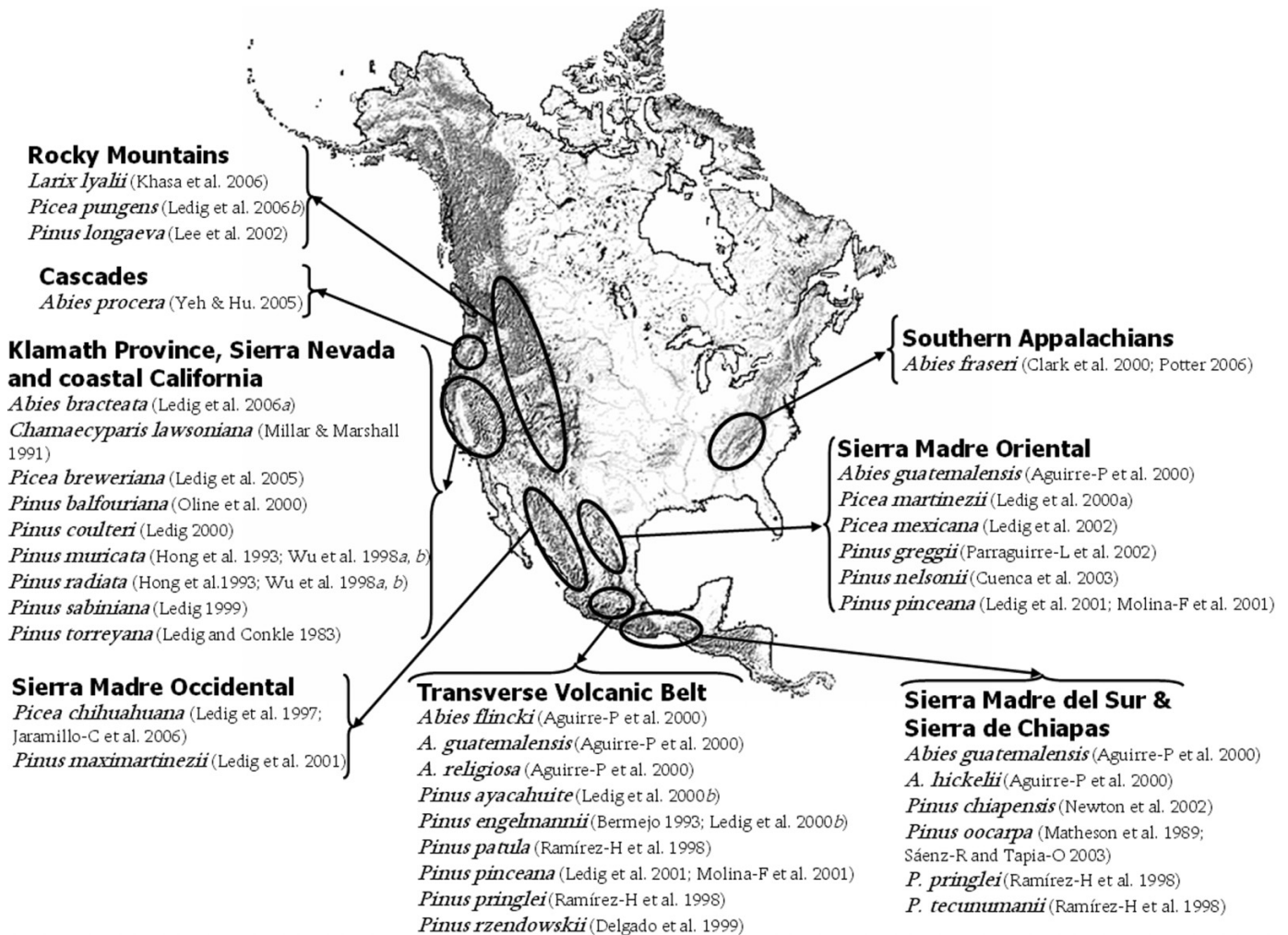


Notwithstanding the fact that secondary contact zones can be aggregated or not in suture zones, their importance from an ecological and evolutionary point of view is undeniable. Introgression may be one of the main sources of adaptive variation in hybridizing species depending on the strength and direction of gene flow and selection against hybrids (reviewed by Barton 2001; Rieseberg et al. 2004). For example, if hybrids are less fit than both parental taxa, the extent of introgression is limited and the reproductive isolation of species is favored, as observed for *Quercus lobata* and *Quercus douglasii* in California (Craft et al. 2002). On the other hand, if some hybrid genotypes are fitter than those of one or both parental species in some particular environments, introgression will make a significant adaptive contribution to the receptive species, such as made evident in the contact zone between *Populus fremontii* and *Populus angustifolia* in the southwestern United States (Martinsen et al. 2001). Moreover, in some extreme cases when hybrids are fitter than both parents in an exclusive environment, hybrid

speciation may occur (e.g., Rieseberg 1997). However, hybrid species seem to arise more easily in annual taxa, which are characterized by rapid chromosomal evolution, short generation times, and significant selfing, than in forest trees (reviewed by Rieseberg 1997; Barton 2001; but see Wang et al. 2001).

Whether this balance between selection and interspecific gene flow can explain, on its own, the long-term maintenance of hybrid zones and the lack of reproductive isolation within many ancient North American genera such as *Abies*, *Picea*, *Pinus*, *Populus*, *Prunus*, and *Quercus* is still a source of debate. The apparent retention of chlorotypes in hybridizing species while in sympatry (e.g., Martinsen et al. 2001; Dodd and Kashani 2003; Jaramillo-Correa et al. 2008) points indeed to a major role of selection and hitchhiking (reviewed by Cruzan and Arnold 1999). However, other factors such as large ancestral population sizes, long generation times, or recurrent long-distance gene flow, which might hamper the fixation of alleles leading to reproductive isola-

Fig. 4. Location of mountain conifer species of North America, which are restricted to putative interglacial refugia and for which population genetic data are available.



tion, also have to be taken into account (reviewed by Petit and Hampe 2006).

Further consequences of introgression and hybridization include the origin of new cytoplasmic variants by genomic recombination (e.g., Marshall et al. 2001; Jaramillo-Correa and Bousquet 2005) and the risk of extinction of rare species by recurrent hybridization with more common relatives or widely planted exotic taxa (reviewed by Wolf et al. 2001). Accordingly, secondary zones of contact of forest tree species should represent a priority for future evolutionary studies and gene conservation programs.

Founding events and bottleneck effects

The colonization of an empty habitat by an expanding species or population is generally driven by two complementary processes: the gradual expansion of a migration front and the long-distance dispersal of a few individuals (e.g., Ibrahim et al. 1996; Petit et al. 1997; Bialozyt et al. 2006). Computer simulations have shown that when recolonization takes place at a slow pace and in the form of an expanding leading front, most of the genetic diversity present in the populations is preserved in the newly established stands (Ibrahim et al. 1996). On the other hand, when

the new populations are established from a few individuals dispersed far ahead of the migration front, they usually exhibit high levels of population differentiation and harbor lower genetic diversity than the core populations from which they originated (Bialozyt et al. 2006). In addition to these founder effects, the newly established stands can suffer further reductions of gene diversity by continued genetic drift if incoming gene flow from the migration front is restricted (Petit et al. 1997) or by the action of the strong selective forces that they might have to face in these newly colonized habitats (e.g., Kirkpatrick and Barton 1997).

In North America, the study of peripheral forest tree populations has revealed contrasting patterns. For example, peripheral stands of *Pinus rigida*, *Pinus contorta*, *Pinus coulteri*, and *Pseudotsuga menziesii* exhibited much lower amounts of nuclear genetic variation than core populations (e.g., Guries and Ledig 1982; Li and Adams 1989; Ledig 2000 among others), a pattern that could be associated with long-distance dispersal events. In *Picea sitchensis* and *Betula occidentalis*, the patterns observed suggested migrating fronts at different speeds. There was a marked although progressive decline of diversity towards the range margins of *Picea sitchensis* (Mimura and Aitken 2007), while this de-

cline was smoother in *Betula occidentalis* (Williams and Arnold 2001). Range limit stands of *Abies procera* (Yeh and Hu 2005), *Alnus rubra* (Hamann et al. 1998), *Picea mariana* (Laberge et al. 2000; Gamache et al. 2003), *Pinus edulis* (Betancourt et al. 1991), *Pinus strobus* (Beaulieu and Simon 1994), and *Populus tremuloides* (Jelinski and Cheliak 1992) harbored as much nuclear genetic diversity as core populations with little population differentiation. Such high and quite uniform amounts of genetic diversity in the marginal stands of these taxa were interpreted as gradual expansion of a migrating front or as a significant exchange of pollen with the core populations (Betancourt et al. 1991; Hamann et al. 1998). However, in some of the particular cases above (e.g., Williams and Arnold 2001; Yeh and Hu 2005), ancient founding events could still be inferred from the significant inbreeding and population differentiation detected in some peripheral stands. The maintenance of high levels of genetic diversity in peripheral stands submitted to extreme climatic conditions and demographic decline is also possible through vegetative maintenance over hundreds and even thousands of years (Laberge et al. 2000; Gamache et al. 2003). In this and other cases, the present genetic diversity is likely relictual in nature and representative of once much larger and less isolated populations probably dating back to the Holocene thermal maximum around 4000 years ago in northeastern Canada (e.g., Terasmae and Anderson 1970; Kaufman et al. 2004).

Future challenges and research prospects

The use of phylogeographic studies to infer the effect of past historical events in the evolution of species has become a common approach during the last 25 years (Petit and Vendramin 2006). However, only the formal comparative analysis of the patterns detected in these studies with those from palynological, paleobotanical, geographic, or geological surveys can assure a more complete and adequate interpretation (Hu et al. 2008; Petit et al. 2008). Such comparative analyses will provide an adequate framework to test for specific hypotheses related to the vicariance factors, the location of cryptic refugia, the description of migration paths, and the responses of species to past and future climate changes (Petit et al. 2008; Provan and Bennett 2008). In Europe, the synergistic collaboration between the paleobotanical and genetics communities has revealed astounding conclusions such as the maintenance of an almost intact genetic structure in forest tree species during millions of years (e.g., Magri et al. 2007). However, in North America, such collaboration is still incipient and limited to a few studies (e.g., Betancourt et al. 1991; Laberge et al. 2000). The complete integration of these complementary disciplines is perhaps the greatest challenge in the near future at both the continental and regional scales.

The implementation of new technologies and detection methods, such as the amplification of ancient or fossil DNA, will allow tackling recurrent issues such as the location of cryptic northern refugia in different parts of the continent (e.g., Bennett and Pauducci 2006). Similarly, the application of genomic approaches and high-throughput genotyping in model forest trees such as *Picea*, *Pinus*, *Populus*, and *Quercus* (reviewed by Brunner et al. 2007) will allow

the detection of adapting forces during the last postglacial expansion (e.g., Mimura and Aitken 2007; Namroud et al. 2008) and the estimation of the adaptability of current populations when facing future climate changes (e.g., Grivet et al. 2008). Genomic approaches will also make it possible to address the role of selection in the long-term maintenance of secondary zones of contact in North America, as hypothesized from different theoretical and empirical works (Barton 2001; Martinsen et al. 2001). The transferability of this information from model taxa to threatened congeneric species will provide valuable information for the development of more efficient conservation programs (e.g., Jaramillo-Correa et al. 2006). In addition, these approaches will be the source of numerous polymorphisms that could be implemented into deep coalescent analyses complemented by fossil or paleobotanical data (e.g., Cruzan and Templeton 2000; Lapointe and Rissler 2005; Morris et al. 2008). This implementation will allow testing for pseudo-congruent patterns in codistributed species such as discerned in the Pacific Northwest (e.g., Carstens et al. 2005), where mesic forest species have developed the same phylogeographic structure and genetic discontinuities for different reasons and at different historical times.

At the biome level, comparative phylogeographic studies will help to understand how co-distributed species have responded to ancient climate changes and will allow the prediction of potential responses to future changes (e.g., Hu et al. 2008; Provan and Bennett 2008). The resolution of these phylogeographic surveys can be additionally improved with the inclusion of interacting taxa, such as trees and their symbionts, pathogens, pollinators, or seed dispersers. To date, the issue of co-localization of genetic lineages has been mainly addressed from an ecological point of view (e.g., Siepielski and Benkman 2004; Smith and Benkman 2007; but see Burban et al. 1999). However, as recently shown for model animal species and for humans (reviewed by Nieberding and Olivieri 2007), the use of new automated and high-throughput genomic and bioinformatics approaches should allow expanding these studies at the phylogeographic level.

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